

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461095.833 +/- 0.041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.102 +/- 0.038
Transit depth (Rp/Rs)^2	1.04 +/- 0.77 [%]
Orbital Inclination (inc)	88.97 +/- 2.69
Airmass coefficient 1 (a1)	1.0141 +/- 0.0069
Airmass coefficient 2 (a2)	-0.011 +/- 0.043
Scatter in the residuals of the lightcurve fit is	0.68 %
Best Comparison Star	#2 - [139, 104]
Optimal Aperture	3.51
Optimal Annulus	4.17
Transit Duration (day)	0.06 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.102 ± 0.038 vs 0.1276 (deviation 0.67σ)
Duration fit vs published	0.06 d vs 0.1179 d (deviation 1.69σ)
Mid-time O–C	11.6 min
Depth SNR	2.7
Residual scatter	0.68 %

Light curve

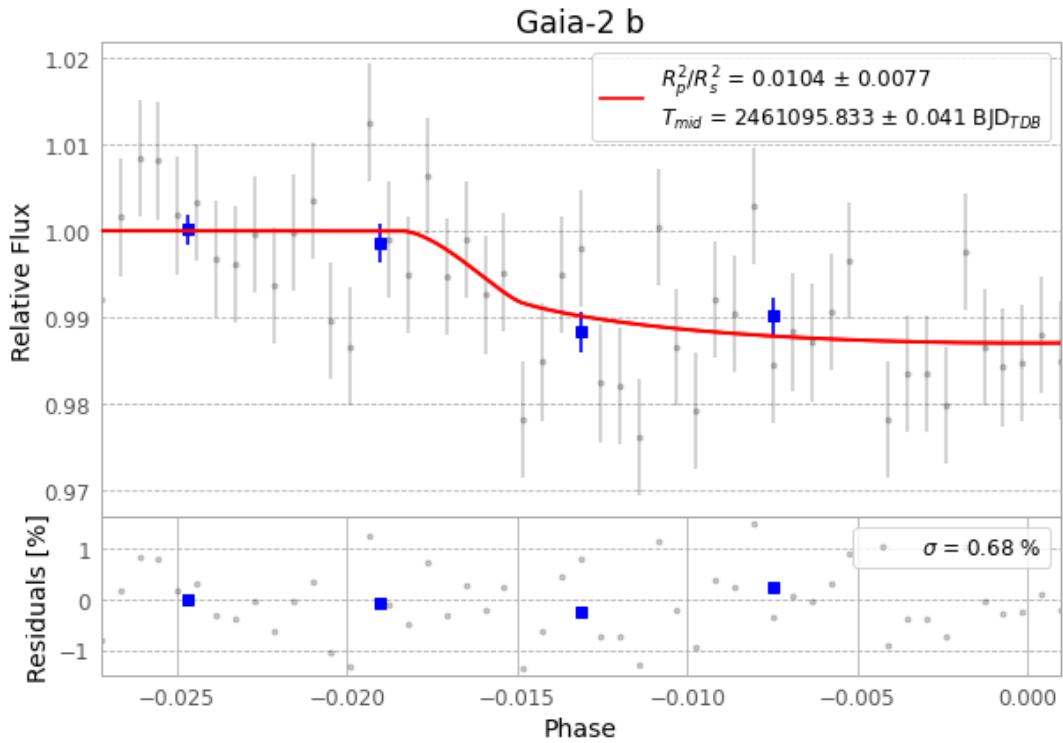


Figure 1 — FinalLightCurve_Gaia-2 b_2026-02-23.png (SHA-256 0ea09b03fc9daa5d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [435, 90], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cd55a32f08a1418f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

51 FITS frames (33 MB), Gaia-2260224053015.FITS → Gaia-2260224080015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Gaia-2-260224.log (dcf2c0c71cc7...), FinalLightCurve_Gaia-2 b_2026-02-23.png (0ea09b03fc9d...), FinalLightCurve_Gaia-2 b_2026-02-23.csv (6cb85a08af8f...)

Compute resources

Wall time 136.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 22:48Z.